The Higgs boson in $H \to ZZ^* \to 4\ell$ with ATLAS Open Data: mass, significance and a classifier

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Abstract

Using the ATLAS Open Data for education (13 TeV, 2015–2016, 36.6 fb^{-1}), I reproduce the official four-lepton notebook exactly, correct its selection and its significance estimate, and extend it in three ways. An unbinned likelihood fit, calibrated and validated with 2000 toy experiments, gives $m_H = 125.06^{+0.93}_{-0.92}$ GeV (statistical only). A profile-likelihood fit of the $m_{4\ell}$ spectrum gives an observed significance of 3.96σ (expected 6.64σ) and a signal strength $\hat{\mu} = 0.72^{+0.22}_{-0.20}$. A classifier on lepton kinematics, tested on events it never saw, does not improve the expected significance. All choices were fixed on simulation before the data were fitted, and every number below is produced by one script, `scripts/make_all.py`.

1 Data, simulation and selection

The Higgs boson was discovered in 2012 by ATLAS and CMS [1, 2], with the four-lepton channel among the most sensitive: its final state is fully reconstructed and its background is small. This analysis uses the ATLAS Open Data release `2025e-13tev-beta` [3], skim `exactly4lep`: 1.36 million data events and simulated samples for the signal ($m_H = 125$ GeV, eight production and decay samples), the irreducible ZZ^* background, and smaller backgrounds (Z+jets, $t\bar{t}$, $t\bar{t}V$, VVV). Each simulated event has the weight

$$w = \frac{L\,\sigma\,\epsilon_{\text{filter}}\,k\,w_{\text{MC}}\,\prod SF}{\sum w_{\text{MC}}}, \qquad L = 36.6\,\text{fb}^{-1}, \tag{1}$$

with the pile-up, electron, muon and trigger scale factors SF. Masses and energies are in GeV.

Reproducing the notebook. The official notebook `HZZAnalysis.ipynb` [4] requires a trigger and a trigger-matched lepton, $p_T > 20, 15, 10$ GeV for the first three leptons, identification and isolation, a lepton-flavour sum compatible with $ee\mu\mu$, $4e$ or 4μ, and zero total charge. It quotes $(S+B)/\sqrt{B + 0.3B^2} = 4.43$ in the window $122.5 < m_{4\ell} < 130$ GeV. My implementation reproduces its stored outputs to machine precision: $S + B = 28.729$, $B = 10.301$, metric 4.426. Three shortcuts appear on the way. The printed "N_{sig}" is signal plus background ($S = 18.428$). The $0.3B^2$ term corresponds to a background uncertainty of $\sqrt{0.3} = 55\%$, not 30%. The p_T thresholds are applied to stored lepton positions, and leptons are stored electrons first, not ordered in p_T. The notebook also takes the absolute value of every weight factor.

Corrected selection. I apply the p_T thresholds to p_T-ordered leptons, use signed weights, and pair the leptons into Z candidates: Z_1 is the same-flavour, opposite-charge pair closest to m_Z, Z_2 the remaining pair, with $50 < m_{12} < 106$ GeV and $m_{34} > 12$ GeV. The pairing rejects charge-zero combinations such as $e^+e^+\mu^-\mu^-$, which pass the notebook's sums; in practice no data event has that topology, and the mass cuts do the work. In the window, the median significance of a counting experiment with known background [5],

$$Z_A = \sqrt{2\,[(s+b)\ln(1+s/b) - s]}, \tag{2}$$

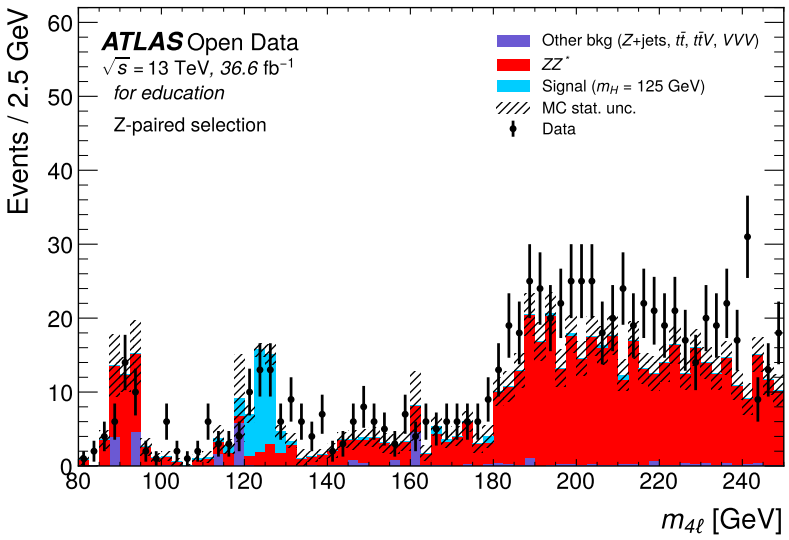


Figure 1: $m_{4\ell}$ after the corrected selection: data and stacked simulation with its statistical uncertainty.

rises from 4.70 (notebook selection) to 4.83 with all eight signal samples, 4.91 with signed weights, 6.42 with ordered p_T cuts, and 7.82 with the Z pairing ($S = 28.8$, $B = 6.6$). Figure 1 shows the spectrum. With signed weights, the data exceed the simulation by 40% at 180–250 GeV, where ZZ^* dominates; the weight normalization was checked against the unskimmed sample, so the deficit is more likely a missing process (no $gg \to ZZ^*$ sample is in the list). This deficit sets the background treatment of Section 3.

2 Higgs boson mass

Model. The $N = 84$ data events in $110 < m_{4\ell} < 140$ GeV are fitted with the extended unbinned likelihood

$$-\ln L(m_H, N_s, N_b) = N_s + N_b - \sum_{i=1}^{N} \ln[N_s\, f_s(m_i;\, m_H + \delta) + N_b\, f_b(m_i)]\,, \tag{3}$$

minimized with MINUIT [6] through iminuit. The signal shape f_s is a double-sided Crystal Ball (a Gaussian core of width $\sigma = 2.08$ GeV with power-law tails), whose width and tail parameters are fixed from a weighted fit to the $H \to ZZ^* \to 4\ell$ simulation with even event numbers. The fitted peak sits $\delta = -0.350 \pm 0.010$ GeV below the generated mass, because of final-state radiation and energy loss, and this offset is subtracted through $f_s(m; m_H + \delta)$. The background f_b is an exponential with its slope fixed from the ZZ^* simulation; a second-order polynomial improved $-2\ln L$ by only 1.2 and was not used. The uncertainty on m_H is the MINOS interval, $-2\Delta\ln L = 1$ with N_s and N_b profiled.

Validation on simulation. 2000 toy experiments at the simulated yields ($N_s = 39.1$, $N_b = 27.6$) all converge and give a pull $(\hat{m}_H - m_H)/\sigma_{\hat{m}_H}$ with mean -0.010 ± 0.023 and width 1.013 ± 0.016: the uncertainty is correctly calibrated. The median expected uncertainty is 0.50 GeV. The odd-event simulation, unused for the shape and the calibration, fitted as weighted pseudo-data, returns $m_H = 125.15 \pm 0.44$ GeV for a generated 125.0 GeV.

Result. The fit to data (Figure 2) gives

$$m_H = 125.06^{+0.93}_{-0.92}\ \text{GeV} \quad \text{(statistical only)}, \tag{4}$$

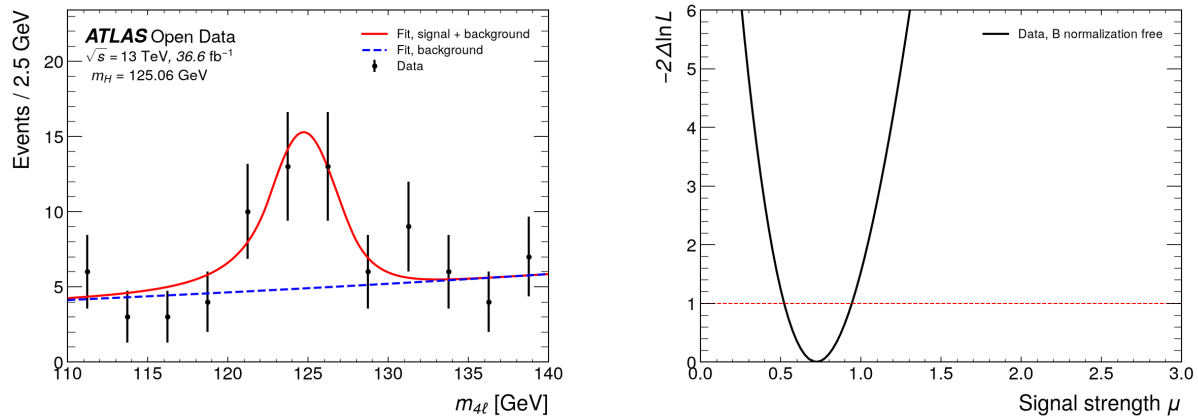


Figure 2: Left: unbinned fit of $m_{4\ell}$ in data, shown in 2.5 GeV bins. Right: profile likelihood of the signal strength μ in data, with the background normalization free.

with a Poisson deviance of 5.9 for 9 degrees of freedom ($p = 0.75$). It agrees with the ATLAS four-lepton measurement on 36.1 fb^{-1} of the same 2015–2016 data, 124.79 ± 0.37 GeV [7], and with the PDG average 125.13 ± 0.11 GeV [8], within a third of my uncertainty. Since the ATLAS result uses the same collisions, the agreement with it is a consistency check, not an independent test. My uncertainty is almost twice the expected 0.50 GeV. Below 130 GeV the data agree with simulation (16 against 16.7 events in 110–120 GeV, 42 against 42.6 in 120–130 GeV), but 130–140 GeV holds 26 events where 7.4 are predicted. With its shape fixed, the background rises everywhere to follow this sideband ($N_b = 59.1 \pm 9.8$, $N_s = 24.9 \pm 7.9$), which widens the likelihood. Only 0.9% of the toys at simulated yields have an error this large; at the fitted yields 30% do. A Gaussian signal shape moves $\hat{m}_H$ by +0.91 GeV on data (spread 0.67 GeV in toys), so the shape choice matters at the level of the statistical error.

3 Significance with a profile likelihood

Model. The $m_{4\ell}$ spectrum in 110–160 GeV (20 bins of 2.5 GeV, 124 data events) is modelled with pyhf [9], in the HistFactory form: in each bin $\nu_i = \mu\,\kappa_L\,\gamma_i\,s_i + \mu_b\,\gamma_i\,b_i$, with the signal strength μ as parameter of interest, a free background normalization μ_b, a luminosity factor κ_L constrained to 0.87% [10], and one factor γ_i per bin for the simulation's statistical uncertainty. The background normalization is free because of the 40% deficit at high mass; this was decided before the fit to data. The test statistic is $q_0 = -2\ln[L(\mu{=}0, \hat{\hat{\theta}})/L(\hat{\mu}, \hat{\theta})]$ and $Z = \sqrt{q_0}$ in the asymptotic approximation [5].

Validation. Single-bin pyhf models reproduce the closed-form Asimov significance, with and without a background uncertainty, within the 10^{-3} relative tolerance of the tests; the background uncertainty is a Poisson-constrained (`shapesys`) term, which is the on/off problem behind the closed-form formula. On the full model, the median of 1000 signal-plus-background toys is 6.50 ± 0.04 against the Asimov 6.64: the asymptotic formula is slightly optimistic. For $s = 10$, $b = 5$, 10^7 counting toys give $p_0 = 2.31 \times 10^{-4}$ against the asymptotic 1.59×10^{-4}.

Result. The observed significance is $Z = 3.96$ ($p_0 = 3.8 \times 10^{-5}$), and $\hat{\mu} = 0.72^{+0.22}_{-0.20}$ (Figure 2, right). The expected significance is 6.64. The gap has two parts: the fitted background is 1.8 times the simulated one (96.6 against 53.7 events), which by itself lowers the expected value to 5.18, and $\hat{\mu}$ is 1.3σ below 1. Both agree with the mass fit. Table 1 shows how much the common shortcuts overstate the significance.

Method	Notebook	Expected	Observed
Notebook metric $(S+B)/\sqrt{B+0.3B^2}$, window	4.43	7.96	–
$S/\sqrt{B}$, window	5.74	11.19	–
Counting Asimov, known B, window	4.70	7.82	–
Counting Asimov, 30% B uncertainty, window	3.08	5.38	–
Shape fit, B fixed to simulation	–	8.74	not valid
Shape fit, B normalization free	–	6.64	**3.96**
Same, 5 GeV bins	–	6.37	4.18

Table 1: Discovery significance by method. "Window" is $122.5 < m_{4\ell} < 130$ GeV; shape fits use 110–160 GeV. Expected and observed use the corrected selection. The notebook column also differs by abs weights and 7 instead of 8 signal samples. The fixed-background fit to data is not valid: the data reject that normalization.

4 Classifier

Method. The classifier separates signal from all backgrounds with ten features: the four lepton p_T, m_{12}, m_{34}, the p_T and rapidity of the four-lepton system, $\cos\theta^*$ of Z_1 in the four-lepton rest frame, and the smallest ΔR between leptons. $m_{4\ell}$ is not a feature, so that the spectrum stays usable for the fit. It is trained on simulation with even event numbers, using absolute weights (negative weights cannot be used as training weights), and evaluated on odd event numbers with signed weights and yields doubled. A score threshold, chosen on the training half, defines two categories, which are fitted together with the model of Section 3, each with its own free background normalization.

A model change, on simulation. The planned gradient-boosted trees [11] overtrained: only 289 background events are available for training, and the train and test background scores disagreed (Kolmogorov–Smirnov $p = 8.6 \times 10^{-14}$, even with small trees). Following a rule fixed before that test, the analysis switched to logistic regression on the same features, which shows no overtraining (KS $p = 0.98$ for signal, 0.43 for background).

Result. The ROC AUC is 0.702 on the test half. The high-score category is purer ($S/B = 2.2$ against 0.7), but the expected significance on the test half is 5.97 with two categories against 6.06 inclusive: no gain. The purer category holds only about five background events, whose free normalization must be fixed by its own sidebands, and the threshold chosen on the training scores is tighter than the test half supports. Although $m_{4\ell}$ is not a feature, the score partly learns it: the background in the high category sits 9 GeV lower in $m_{4\ell}$, which a χ^2 shape test with so few events does not detect ($p = 0.37$). On data, with the classifier and threshold frozen, the two-category fit gives $Z = 3.29$ against 3.90 inclusive with the same templates. The inclusive result of Section 3 remains the main one.

5 Limitations

- The data exceed the simulation by 40% at high mass and by a factor 3.5 at 130–140 GeV; neither cause is identified. The free background normalization absorbs the first, but the background shape is taken from simulation.
- Few systematic uncertainties: luminosity, a free background normalization and the simulation's statistical uncertainty in the significance; none on the mass (no lepton energy or momentum scale variations). The signal-shape choice alone moves m_H by up to 0.9 GeV.
- The background simulation is sparse: one Z+jets event of weight 5.8 is 11% of the background in 110–160 GeV, and 289 background events are all the classifier can learn from.

- The selection is simplified: no $m_{4\ell}$-dependent m_{34} cut, no ΔR cut, no J/ψ veto. The mass calibration comes from a single simulated mass point.
- Blinding is partial: the data were plotted at the selection stage; every later choice was fixed on simulation before the corresponding fit to data.

6 Conclusion

With $36.6\,\text{fb}^{-1}$ of open 13 TeV data, the four-lepton channel gives $m_H = 125.06^{+0.93}_{-0.92}\,\text{GeV}$ and an observed significance of 3.96σ, with methods that were validated on simulation and toys before being applied to data. The main lesson is that honest statistics change the answer: in the corrected selection, a naive $S/\sqrt{B}$ of 11.2 becomes an expected 6.64 and an observed 3.96 once the background normalization is constrained by the data instead of assumed. Better background simulation, not a better classifier, is what this analysis needs most.

References

[1] ATLAS Collaboration, Observation of a new particle in the search for the Standard Model Higgs boson with the ATLAS detector at the LHC, Phys. Lett. B 716 (2012) 1, arXiv:1207.7214.

[2] CMS Collaboration, Observation of a new boson at a mass of 125 GeV with the CMS experiment at the LHC, Phys. Lett. B 716 (2012) 30, arXiv:1207.7235.

[3] ATLAS Collaboration, ROOT ntuple format 2015–2016 proton-proton Open Data for Education and Outreach beta release from the ATLAS experiment, CERN Open Data Portal (2025), doi:10.7483/OPENDATA.ATLAS.B5M9.44TN.

[4] ATLAS Open Data, `HZZAnalysis.ipynb`, `https://github.com/atlas-outreach-data-tools/notebooks-collection-opendata`, directory `13-TeV-examples/uproot_python`.

[5] G. Cowan, K. Cranmer, E. Gross, O. Vitells, Asymptotic formulae for likelihood-based tests of new physics, Eur. Phys. J. C 71 (2011) 1554, arXiv:1007.1727.

[6] F. James, M. Roos, Minuit: a system for function minimization and analysis of the parameter errors and correlations, Comput. Phys. Commun. 10 (1975) 343.

[7] ATLAS Collaboration, Measurement of the Higgs boson mass in the $H \to ZZ^* \to 4\ell$ and $H \to \gamma\gamma$ channels with $\sqrt{s} = 13\,\text{TeV}$ pp collisions using the ATLAS detector, Phys. Lett. B 784 (2018) 345, arXiv:1806.00242.

[8] F. Takahashi et al. (Particle Data Group), Review of Particle Physics, Int. J. Mod. Phys. A 41 (2026) 2630011.

[9] L. Heinrich, M. Feickert, G. Stark, K. Cranmer, pyhf: pure-Python implementation of HistFactory statistical models, J. Open Source Softw. 6 (2021) 2823.

[10] ATLAS Collaboration, Luminosity determination in pp collisions at $\sqrt{s} = 13\,\text{TeV}$ using the ATLAS detector at the LHC, Eur. Phys. J. C 83 (2023) 982, arXiv:2212.09379.

[11] F. Pedregosa et al., Scikit-learn: machine learning in Python, J. Mach. Learn. Res. 12 (2011) 2825.